



Acute Heart Failure

(Congestive Heart Failure)

Full Review: Oct 2025 By [Nowell M. Fine](#), MD, SM, Libin Cardiovascular Institute, Cumming School of Medicine, University of Calgary | Peer reviewed by [Jonathan G. Howlett](#), MD, Cumming School of Medicine, University of Calgary
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Heart failure (HF) is a clinical syndrome in which the heart is not able to meet the metabolic demands of the body due to a structural and or functional cardiac abnormality, leading to low cardiac output, elevated ventricular filling pressure, or both. Acute heart failure includes both new-onset heart failure and acute decompensated heart failure (an exacerbation of chronic heart failure). In both, there are progressive signs or symptoms of cardiac congestion, low output, or both.. Timely medical intervention, directed at reducing symptoms and managing the cause, is necessary.

(See also [Overview of Heart Failure](#).)

Acute heart failure refers not only to the timing of clinical presentation within the overall disease course, but the importance of timely medical intervention ([1](#)). It may include new-onset (de novo) heart failure or acute decompensation of previously diagnosed heart failure. Acute heart failure is classified phenotypically and further risk-stratified based on the overall severity of illness, alteration in perfusion ("warm" versus "cold"), the presence of congestion ("dry" versus "wet"), and other details of clinical presentation ([2](#), [3](#)).

General references

1. [Mebazaa A, Yilmaz MB, Levy P, et al.](#) Recommendations on pre-hospital & early hospital management of acute heart failure: a consensus paper from the Heart Failure Association of the European Society of Cardiology, the European Society of Emergency Medicine and the Society of Academic Emergency Medicine. *Eur J Heart Fail.* 2015;17(6):544-558. doi:10.1002/ejhf.289
2. [Arrigo M, Jessup M, Mullens W, et al.](#) Acute heart failure. *Nat Rev Dis Primers.* 2020;6(1):16. Published 2020 Mar 5. doi:10.1038/s41572-020-0151-7
3. [Hollenberg SM, Warner Stevenson L, Ahmad T, et al.](#) 2019 ACC Expert Consensus Decision Pathway on Risk Assessment, Management, and Clinical Trajectory of Patients Hospitalized With Heart Failure: A Report of the American College of Cardiology Solution Set Oversight Committee [published correction appears in *J Am Coll Cardiol.* 2020 Jan 7;75(1):132. doi:

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doi:10.1016/j.jacc.2019.08.001

Symptoms and Signs of Acute Heart Failure

Manifestations of heart failure differ depending on the extent to which the left ventricle (LV) and right ventricle (RV) are initially affected. Acute RV failure is discussed separately.

History

In LV failure, the most common symptoms are dyspnea and fatigue due to increased pulmonary venous pressures with or without [pulmonary edema](#), and low cardiac output (at rest or inability to augment cardiac output during exertion). Dyspnea may be exertional or occur with rest, depending on the severity of disease. [Cardiogenic shock](#), if present, may cause abnormalities in mental status and organ system function.

Examination

General examination may detect signs of systemic or cardiac disorders that cause or aggravate heart failure (eg, [anemia](#), [hyperthyroidism](#), [alcohol use disorder](#), [hemochromatosis](#), [atrial fibrillation](#) with rapid rate, [mitral regurgitation](#)).

In LV failure, tachycardia and tachypnea may occur. Heart rate is generally increased as a compensatory mechanism, but bradycardia may occur during decompensation or if there is an underlying bradyarrhythmia. Blood pressure may be high, normal, or low, depending on the changes in cardiac output, volume status, systemic vascular resistance due to the underlying disease, and the compensatory response. Patients with severe LV failure may appear visibly dyspneic or cyanotic, hypotensive, and confused or agitated because of hypoxia and poor cerebral perfusion. Some of these less specific symptoms (eg, confusion) are more common in older patients.

Central cyanosis (affecting all of the body, including warm areas such as the tongue and mucous membranes) reflects severe hypoxemia. Peripheral cyanosis of the lips, fingers, and toes reflects low blood flow with increased oxygen extraction. If vigorous massage improves nail bed color, cyanosis may be peripheral; increasing local blood flow does not improve color if cyanosis is central.

[Cardiac findings](#) in acute heart failure with reduced ejection fraction (HFrEF) include:

- Diffuse, sustained, and laterally displaced apical impulse
- Audible and occasionally palpable third (S3) and fourth (S4) heart sounds
- Accentuated pulmonic component (P2) of the second heart sound (S2)

These abnormal heart sounds also can occur in heart failure with preserved ejection fraction (HFpEF) or heart failure with minimally reduced ejection fraction (HFmrEF). A pansystolic murmur of mitral regurgitation at the apex may occur, particularly if the LV is significantly dilated. Jugular venous distention indicates elevated right atrial pressure. Pulsus alternans (an pulse alternating between strong

and weak beats) may accompany severely reduced LV systolic function and has historically been considered a poor prognostic sign ([1](#), [2](#)).

Pulmonary findings include early inspiratory basilar crackles that do not clear with coughing and, if pleural effusion is present, dullness to percussion and diminished breath sounds at the lung base(s). Wheezing or stridor may be present due to inflammation of smaller airways as a result of pulmonary edema, or due to compression of larger airways by enlarged cardiac structures (ie, the left atrium or great vessels).

Symptoms and signs references

1. [Aslanger E, Aggöl B, Albayrak DG](#). Pulsus Alternans: Caught in Action. *Catheter Cardiovasc Interv*. 2025;105(1):270-271. doi:10.1002/ccd.31295
2. [Schaefer S, Malloy CR, Schmitz JM, Dehmer GJ](#). Clinical and hemodynamic characteristics of patients with inducible pulsus alternans. *Am Heart J*. 1988;115(6):1251-1257. doi:10.1016/0002-8703(88)90017-8

Diagnosis of Acute Heart Failure

- History and physical examination
- Chest radiograph
- Echocardiography
- Laboratory evaluation, including neurohormonal biomarkers such as brain (B-type) natriuretic peptide (BNP) or N-terminal-pro-BNP (NT-pro-BNP) levels
- Sometimes ECG, cardiac radionuclide scan, cardiac MRI, cardiac catheterization, and other tests for etiology and to assess effect on end-organ function

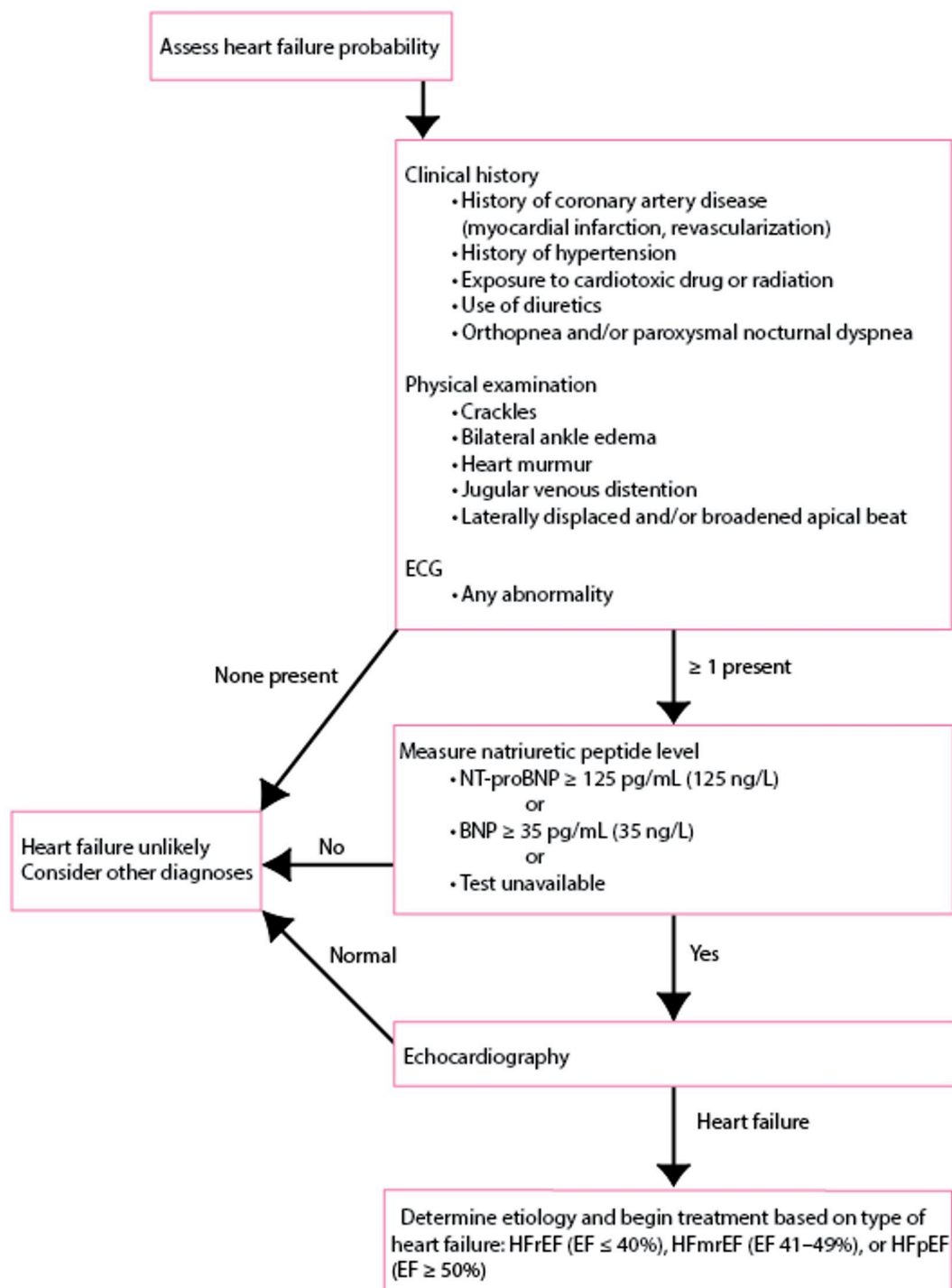
Clinical findings (eg, exertional dyspnea or fatigue, orthopnea, edema, tachycardia, pulmonary crackles, S3, jugular venous distention) suggest heart failure but are usually not apparent early. Some similar symptoms may result from [chronic obstructive pulmonary disease \(COPD\)](#) or recurrent [pneumonia](#) or may be erroneously attributed to obesity or old age. Suspicion for heart failure should be high in patients with a history of [myocardial infarction](#), [hypertension](#), [valvular disorders](#), or murmurs and should be moderate in any patient who is older or has [diabetes](#).

Chest radiograph, ECG, and an objective test of cardiac function, typically echocardiography, should be done (see figure [Diagnosis of Heart Failure of Acute Onset](#)). Blood tests, except for BNP levels, are not used for diagnosis but are useful for identifying cause and systemic effects ([1](#), [2](#)). Similarly, blood tests, including lactic acid, electrolytes, renal function, and hepatic function, may identify effects on tissue and organ perfusion.

Diagnosis of Acute Heart Failure

Data from [McDonagh TA, Metra M, Adamo M, et al](#): 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure: Developed by the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC) with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur Heart J* 42(36):3599-3726, 2021. doi: 10.1093/eurheartj/ehab368.

ECG = electrocardiography; EF = ejection fraction; NT-pro-BNP = N-terminal-pro-BNP; BNP = Brain (B-type) natriuretic peptide; HFmrEF = heart failure with mildly reduced ejection fraction; HFpEF = heart failure with preserved ejection fraction; HFrEF = heart failure with reduced ejection fraction.



Chest radiograph

Chest radiograph findings suggesting heart failure include an enlarged cardiac silhouette, pleural effusion, fluid in the major fissure, and horizontal lines in the periphery of lower posterior lung fields (Kerley B lines). These findings reflect chronic elevation of left atrial pressure and chronic thickening of the intralobular septa due to edema. Upper lobe pulmonary venous congestion and interstitial or alveolar edema may also be present. Careful examination of the cardiac silhouette on a lateral

projection can identify specific ventricular and atrial chamber enlargement. The radiograph may also suggest alternative diagnoses (eg, [COPD](#), [pneumonia](#), [idiopathic pulmonary fibrosis](#), [lung cancer](#)).

ECG

ECG findings are not diagnostic, but an abnormal ECG, especially one showing previous myocardial infarction, left ventricular hypertrophy, left bundle branch block, or tachyarrhythmia (eg, rapid [atrial fibrillation](#)), increases suspicion for heart failure and may help identify the cause. Left or right atrial enlargement may be present. Left ventricular hypertrophy may reflect LV dilation, or voltages may be reduced in the case of severe LV systolic dysfunction.

Other cardiac imaging

Echocardiography can help evaluate chamber dimensions, valve function, LVEF, wall motion abnormalities, LV hypertrophy, diastolic function, pulmonary artery pressure, LV and RV filling pressures, RV function, and pericardial effusion. Intracardiac thrombi, tumors, and calcifications within the heart valves, mitral annulus, and aortic wall abnormalities can be detected. Localized or segmental wall motion abnormalities strongly suggest underlying [coronary artery disease](#) but can also be present with patchy [myocarditis](#).

Doppler or color Doppler echocardiography accurately detects valvular disorders and shunts. The combination of Doppler evaluation of mitral inflow with tissue Doppler imaging of the mitral annulus can help identify and quantify LV diastolic dysfunction and LV filling pressures.

Measuring LVEF can distinguish between predominant HFpEF (EF $\geq$ 50%) and HFrEF (EF $\leq$ 40%). It is important to re-emphasize that heart failure can occur with a normal LVEF.

Speckle-tracking echocardiography, a measure of strain, is useful in detecting subclinical systolic dysfunction and specific patterns of myocardial dysfunction and may provide useful information regarding the pattern of myocardial involvement of cardiomyopathy, and therefore information about the cause.

Radionuclide imaging also can help assess systolic and diastolic function, previous myocardial infarction, and inducible ischemia or myocardial hibernation. It is used most commonly to assess the presence and/or severity of ischemic heart disease and can also be used to quantify left ventricular ejection fraction.

Cardiac MRI using late gadolinium enhancement imaging (LGE, also called fibrosis or scar imaging) is useful to evaluate the cause of myocardial disease and to detect focal and diffuse myocardial fibrosis. Cardiac amyloidosis, sarcoidosis, hemochromatosis, and myocarditis are causes of HF that can be detected with or suspected based on cardiac MRI findings. Patients with acute heart failure must be stabilized in order to undergo cardiac MRI, because the test requires a prolonged duration in a supine position and coordinated breath holding (or, in some cases, general anesthesia with ventilatory support).

Blood tests

Serum BNP levels are often high in heart failure; this finding may help when clinical findings are unclear or other diagnoses (eg, COPD) need to be excluded. It may be particularly useful for patients with a history of both pulmonary and cardiac disorders. NT-pro-BNP, an inactive moiety created when pro-BNP is cleaved, can be used similarly to BNP. However, *a normal BNP level does not exclude the diagnosis of heart failure*, particularly in patients with HFpEF and/or obesity. In HFpEF, BNP levels tend to be approximately 50% of those associated with HFrEF (at similar degrees of symptoms), and up to 30% of patients with acute HFpEF have a BNP level below the commonly used threshold of 100 pg/mL (100 ng/L). Obesity, a common comorbidity in HF, is associated with reduced BNP production and increased BNP clearance, resulting in lower levels.

Besides BNP, recommended blood tests include complete blood count, creatinine, blood urea nitrogen (BUN), electrolytes (including magnesium and calcium), glucose, albumin, ferritin, and liver tests. Thyroid function tests are recommended for patients with atrial fibrillation and for selected, especially older, patients.

Other tests

Thoracic ultrasound is a noninvasive method of detecting pulmonary congestion in patients with heart failure. Sonographic "comet tail artifact" on thoracic ultrasound corresponds to the radiographic finding of Kerley B lines.

Coronary angiography or **CT coronary angiography** is indicated when [coronary artery disease](#) is suspected or the etiology of HF is uncertain.

Cardiac catheterization with intracardiac pressure measurements (invasive hemodynamic assessment) may be helpful in the diagnosis of restrictive cardiomyopathies and constrictive pericarditis. Invasive hemodynamic measurements are also very helpful when the diagnosis of HF is equivocal, particularly in patients with HFpEF. In addition, perturbing the cardiovascular system (eg, exercise testing, volume challenge, medication challenges [eg, [nitroglycerin](#), [nitroprusside](#)]) can be very helpful during invasive hemodynamic testing to help diagnose HF.

Endomyocardial biopsy is sometimes done when an infiltrative cardiomyopathy or acute giant cell myocarditis is strongly suspected but cannot be confirmed with noninvasive imaging (eg, cardiac MRI).

Diagnosis references

1. [McDonagh TA, Metra M, Adamo M, et al](#): 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure: Developed by the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC) with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur Heart J* 42(36):3599-3726, 2021. doi: 10.1093/eurheartj/ehab368
2. [Heidenreich PA, Bozkurt B, Aguilar D, et al](#): 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure: A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation* 145:e876–e894, 2022, doi: 10.1161/CIR.0000000000001062

Treatment of Acute Heart Failure

- Pharmacotherapy
- Respiratory and other organ system support as necessary
- Identification and treatment of cause
- Sometimes device therapy (eg, implantable cardioverter-defibrillator, cardiac resynchronization therapy, mechanical circulatory support)
- Sometimes cardiac transplantation
- Multidisciplinary care

Immediate inpatient treatment is required for patients with acute or worsening heart failure (HF) due to certain disorders (eg, [acute myocardial infarction](#), [atrial fibrillation](#) with a very rapid ventricular rate, severe [hypertension](#), acute [valvular regurgitation](#)), as well as for patients with [pulmonary edema](#), severe symptoms, new-onset HF, or HF unresponsive to outpatient treatment. Patients with mild exacerbations of previously diagnosed HF can be treated at home.

The primary goal is to diagnose and to correct or treat the disorder that led to heart failure.

Acute treatment goals include relieving symptoms and improving hemodynamics (optimizing preload, balancing ventricular function with myocardial oxygen demand, and appropriately reducing afterload); avoiding [hypokalemia](#), renal dysfunction, and symptomatic hypotension; and correcting neurohumoral activation.

The mainstay of the **pharmacologic management** of acute heart failure is decongestive therapy with diuretics, namely loop diuretics (eg, [furosemide](#), [torsemide](#), [bumetanide](#)) in either oral or intravenous form. (See [Pulmonary Edema](#) for a more detailed discussion.) Loop diuretics may be augmented by adding other diuretic classes such as thiazides (eg, [metolazone](#)). Afterload reduction using both oral and intravenous arterial vasodilators (eg, [nitroprusside](#)) can also be used to improve acute heart failure. Pulmonary vasodilators (eg, nitrate therapies) may also be used to reduce symptoms of congestion acutely. Low output heart failure, particularly in the setting of cardiogenic shock, often requires acute treatment with intravenous inotropic agents (eg, [milrinone](#), [dobutamine](#)). Management of the cause of heart failure may also be indicated acutely, particularly for certain causes such as acute ischemia or arrhythmia.

For heart failure with reduced ejection fraction (HFrEF) in the acute setting, guideline-directed medical therapy with a sodium-glucose co-transporter 2 (SGLT2) inhibitor, beta-blocker, angiotensin receptor/neprilysin inhibitor, and mineralocorticoid receptor antagonist should be initiated once the patient is stabilized ([1](#), [2](#)). Medication doses are increased rapidly to target doses during the hospitalization and the first 6 weeks of follow-up to reduce the risk of both mortality and re-admission. See [Chronic Heart Failure](#) and [Medications for Heart Failure](#) for more detailed information on these medications.

Other medications, such as nesiritide (a brain natriuretic peptide), levosimendan (a calcium-sensitizing inotrope), vesnarinone (an inotrope with multiple mechanisms), and ibopamine (a dopaminergic and

adrenergic receptor agonist), may provide short-term symptomatic or hemodynamic improvement but do not improve, and in some cases may worsen, long-term outcomes compared to standard therapy (in randomized trials mortality was higher with vesnarinone and ibopamine and similar with nesiritide ([3](#), [4](#), [5,6](#)). Serelaxin (a recombinant form of the human pregnancy hormone relaxin-2) has been studied, but benefits were not shown in a large international randomized study ([7](#)). Omecamtiv mecarbil (an oral cardiac myosin activator) reduced a composite endpoint of heart failure event or death but did not reduce overall cardiovascular mortality in a large randomized trial. ([8](#)).

See [Medications for Heart Failure](#) for more detailed information on specific medication classes.

Positive airway pressure via high flow nasal cannula, continuous positive airway pressure, or bi-level positive airway pressure, or endotracheal intubation generally reduce left ventricular afterload by increasing intrathoracic pressure and reducing transmural LV pressure.

Arrhythmia treatment

Because arrhythmias can worsen heart failure, it is important to identify and treat the cause of any [arrhythmia](#).

- Electrolytes are normalized.
- Atrial and ventricular rates are controlled.
- Sometimes antiarrhythmic medications are given.

Sinus tachycardia, a common compensatory change in heart failure, usually subsides when HF treatment is effective. If it does not, associated causes (eg, [hyperthyroidism](#), [pulmonary emboli](#), fever, anemia, pain) should be sought. If sinus tachycardia persists despite correction of causes, a beta-blocker, given in gradually increasing doses, may help selected patients. However, lowering heart rate with a beta-blocker can be detrimental to patients with advanced HFpEF (eg, [restrictive cardiomyopathy](#)), in whom stroke volume is fixed because of severe diastolic dysfunction. In these patients, cardiac output is heart rate-dependent, and lowering heart rate can thus lower cardiac output at rest with exertion, or both.

[Atrial fibrillation](#) with an uncontrolled ventricular rate must be treated; the target resting ventricular rate is typically < 80 beats/minute. Beta-blockers are the treatment of choice, although rate-limiting calcium channel blockers may be used cautiously if systolic function is preserved. Adding [digoxin](#), low-dose [amiodarone](#), or other rhythm and/or rate controlling medications may help some patients. Routine conversion to and maintenance of sinus rhythm has not been shown to be superior to rate control alone in randomized trials ([9](#)). However, it is best to make this determination on a case-by-case basis because some patients improve significantly with restoration of normal sinus rhythm. If rapid atrial fibrillation does not respond to medication, permanent pacemaker insertion with complete or partial ablation of the atrioventricular node, or other [atrial fibrillation ablation procedures](#), may be considered to restore a sinus or regular rhythm in selected patients.

Isolated [ventricular premature beats](#), which are common in HF, do not require specific treatment, although rarely very frequent ventricular premature beats (> 15,000/day) have been shown to precipitate heart failure (that reverses with suppression). However, optimization of HF treatments and

correction of electrolyte abnormalities (especially of potassium and magnesium) reduce the risk of ventricular arrhythmias.

Sustained [ventricular tachycardia](#) that persists despite correction of cause (eg, low potassium or magnesium, ischemia) and optimal medical treatment of HF may require an antiarrhythmic medication. [Amiodarone](#), beta-blockers, and [dofetilide](#) are the medications of choice because other antiarrhythmics have adverse proarrhythmic effects when LV systolic dysfunction is present.

Because amiodarone increases [digoxin](#) and [warfarin](#) levels, digoxin and warfarin doses should be decreased by half or stopped. Serum digoxin level and INR (international normalized ratio) level should be routinely monitored. However, medication toxicity can occur even at therapeutic levels. Because long-term use of amiodarone can cause adverse effects, a low dose is used when possible; blood tests for liver function and thyroid-stimulating hormone are done every 6 months. If the chest radiograph is abnormal or dyspnea worsens significantly, chest radiographs and pulmonary function tests are done yearly to check for pulmonary fibrosis. For sustained ventricular arrhythmias, amiodarone may be required; to reduce risk of sudden death, an oral loading dose is given twice a day for 1 to 3 weeks until rhythm control is adequate, then dose is decreased over 1 month to a once daily maintenance dose.

Device therapy and mechanical circulatory support

Device therapy, including [implantable cardioverter-defibrillator](#) (ICD) and [cardiac resynchronization therapy](#) (CRT), is primarily used for chronic heart failure. In the acute setting, however, CRT may provide symptom management, and both CRT and ICD may be used in the management of a refractory arrhythmia causing or exacerbating the heart failure.

Ultrafiltration (venovenous filtration) can be useful in selected hospitalized patients with severe cardiorenal syndrome and volume overload refractory to diuretics. However, ultrafiltration should not be used routinely because evidence does not clearly show long-term clinical benefit ([10](#)).

An **intra-aortic counterpulsation balloon pump** (IABP) is helpful in selected patients with acute HF who have a good chance of recovery (eg, acute HF following myocardial infarction) or in those who need a bridge to a more permanent solution such as cardiac surgery (eg, to fix severe valvular disease or to revascularize multivessel coronary artery disease), an [LV assist device](#), or [heart transplantation](#).

Other forms of temporary mechanical circulatory support for patients with acute HF and cardiogenic shock include surgically placed devices such as **extracorporeal membrane oxygenation** (ECMO, typically with venoarterial cannulation in patient's with heart failure) and **centrifugal flow ventricular assist devices** that can support either the LV, the RV, or both and can also be combined with an oxygenator to provide full cardiopulmonary support. Percutaneously placed devices such as **intravascular microaxial ventricular assist devices** are available for both LV and RV support. Selection of temporary mechanical circulatory support devices is based mainly on availability and local medical center experience.

Please see [device therapy for chronic heart failure](#) for more information.

Treatment references

Prognosis for Acute Heart Failure

Generally, patients hospitalized for acute heart failure have a poor prognosis unless the cause is correctable. Overall combined 5-year survival is estimated at 34%, and 10-year survival at 17%, for patients with after an initial hospitalization for all types of heart failure ([1](#)).

Prognosis reference

1. [Hariharaputhiran S, Peng Y, Ngo L, et al.](#) Long-term survival and life expectancy following an acute heart failure hospitalization in Australia and New Zealand. *Eur J Heart Fail.* 2022;24(9):1519-1528. doi:10.1002/ejhf.2595

Key Points

- Consider acute heart failure in patients with worsening or new onset of exertional dyspnea or fatigue, orthopnea, and/or edema, particularly in those with a history of myocardial infarction, hypertension, or valvular disorders or murmurs.
- Perform chest radiograph, ECG, brain (B-type) natriuretic peptide levels, echocardiography, and additional testing as necessary.
- Treat with guideline-directed medical therapy appropriate to heart failure type prior to hospital discharge.
- Treatment includes control of underlying disorders, medications, sometimes implantable devices (cardiac resynchronization therapy, implantable cardioverter-defibrillators), education and lifestyle changes.

Drug Information for the Topic



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Chronic Heart Failure

(Congestive Heart Failure)

Full Review: Oct 2025 By [Nowell M. Fine](#), MD, SM, Libin Cardiovascular Institute, Cumming School of Medicine, University of Calgary | Peer reviewed by [Jonathan G. Howlett](#), MD, Cumming School of Medicine, University of Calgary
Last updated: Jul 2026

Heart failure (HF) is a clinical syndrome in which the heart is not able to meet the demands of the body, manifested by low cardiac output, elevated ventricular filling pressure, or both. Chronic heart failure refers to the long-term state of this clinical syndrome. Chronic heart failure may be compensated, indicating a relatively stable state that may or may not be symptomatic. When chronic heart failure involves an acute or subacute worsening of symptoms or hemodynamics, it becomes acute decompensated heart failure. Diagnosis is by examination, blood tests, and imaging. Treatment focuses on management of lifestyle and risk factors and guideline-directed medical therapy.

In addition to the 6.7 million people in the United States with heart failure, about one-third of the adult population is at risk for developing heart failure, and about 1 in 4 people will develop it at some point in their lifetime ([1](#)). The definition, pathophysiology, and causes of heart failure, including chronic heart failure, are reviewed in [Overview of Heart Failure](#).

Chronic heart failure is staged as follows ([2](#)):

- A: Risk of HF but no structural or functional cardiac abnormalities or symptoms
- B: Structural or functional cardiac abnormalities but no symptoms
- C: Structural heart disease with symptoms
- D: Refractory HF requiring advanced therapies or palliative care

Patient functional status in stages B to D can be further classified based on symptoms using the [New York Heart Association Functional Classification system](#). The type of heart failure can be further [classified based on the left ventricular ejection fraction](#).

General references

1. [Bozkurt B, Ahmad T, Alexander K, et al](#). HF STATS 2024: Heart Failure Epidemiology and Outcomes Statistics An Updated 2024 Report from the Heart Failure Society of America. *J Card Fail*. 2025;31(1):66-116. doi:10.1016/j.cardfail.2024.07.001

2. [Bozkurt B, Coats AJ, Tsutsui H, et al](#). Universal Definition and Classification of Heart Failure: A Report of the Heart Failure Society of America, Heart Failure Association of the European Society of Cardiology, Japanese Heart Failure Society and Writing Committee of the Universal Definition of Heart Failure. *J Card Fail*. Published online March 1, 2021. doi:10.1016/j.cardfail.2021.01.022

Symptoms and Signs of Chronic Heart Failure

Common chronic heart failure symptoms include exertional dyspnea, exercise intolerance, and peripheral edema (particularly in the ankles and feet) ([1](#)). Other symptoms may include anorexia or early satiety, nausea, presyncope or syncope, and coughing or wheezing (part of cardiac asthma).

In earlier or milder stages, dyspnea occurs with varying degrees of exertion. As HF worsens, dyspnea can occur during rest and at night, sometimes causing nocturnal cough. Dyspnea occurring immediately or soon after lying flat and relieved promptly by sitting up (orthopnea) is common as heart failure advances. In paroxysmal nocturnal dyspnea, dyspnea awakens patients several hours after they lie down and is relieved only after they sit up for 15 to 20 minutes. In severe HF, periodic cycling of breathing ([Cheyne-Stokes respiration](#)—from a brief period of apnea, patients breathe progressively faster and deeper, then slower and shallower until they become apneic and repeat the cycle)—can occur during the day or night; the sudden hyperpneic phase may awaken the patient from sleep. Cheyne-Stokes breathing differs from paroxysmal nocturnal dyspnea in that the hyperpneic phase is short, lasting only 10 to 15 seconds, but the cycle recurs regularly, lasting 30 seconds to 2 minutes. Paroxysmal nocturnal dyspnea is associated with pulmonary congestion, and Cheyne-Stokes respiration with low cardiac output.

Severely reduced cerebral blood flow and hypoxemia can cause chronic irritability and impair mental performance.

Sleep-related breathing disorders, such as [sleep apnea](#) (central and obstructive), are common in HF and may both aggravate HF and be worsened by HF.

[Physical signs](#) specific to heart failure included jugular venous distention or elevated jugular venous pulsations, a third heart sound (S3) or summation gallop, displaced apical impulse due to cardiomegaly, and the hepatojugular reflux ([1](#)). Signs of pulmonary edema (crackles, hypoxemia), poor perfusion (cool extremities, delayed capillary refill), decreased stroke volume (narrow pulse pressure), or a compensatory response (tachycardia, hypertension) may also occur. Weight loss and cardiac cachexia may occur in advanced heart failure.

Symptoms and signs reference

1. [Bozkurt B, Coats AJ, Tsutsui H, et al](#). Universal Definition and Classification of Heart Failure: A Report of the Heart Failure Society of America, Heart Failure Association of the European Society of Cardiology, Japanese Heart Failure Society and Writing Committee of the Universal Definition of Heart Failure. *J Card Fail*. Published online March 1, 2021. doi:10.1016/j.cardfail.2021.01.022

Diagnosis of Chronic Heart Failure

Diagnostic techniques in heart failure are discussed in detail under [Acute Heart Failure](#). Attention to symptoms, weight, serial imaging (eg, echocardiography) and measurement of natriuretic peptides and other biomarkers can help guide treatment and identify acute decompensation.

Treatment of Chronic Heart Failure

- Diet and lifestyle changes
- Treatment of cause
- Medication therapy
- Multidisciplinary care
- Sometimes device therapy (eg, implantable cardioverter-defibrillator, cardiac resynchronization therapy, mechanical circulatory support)
- Sometimes cardiac transplantation

Treatment goals for chronic heart failure include correcting [hypertension](#), preventing [myocardial infarction](#) and progression of [atherosclerosis](#), improving cardiac function, reducing hospitalizations, improving quality of life, and lengthening survival.

Treatment involves dietary and lifestyle changes, [medications](#), devices, and sometimes percutaneous coronary interventions or surgery.

Treatment is multidisciplinary and tailored to the patient, considering causes, symptoms, and response to medications, including adverse effects. There are several evidence-based therapies for chronic [heart failure with reduced ejection fraction](#) (HFrEF) ([1](#), [2](#)). There are fewer evidence-based treatments for chronic [heart failure with preserved ejection fraction](#) (HFpEF), chronic [heart failure with mildly reduced ejection fraction](#) and improved ejection fraction (HFmrEF and HRimpEF), and chronic [right ventricular failure](#) ([3](#)).

Medications

See [Medications for Heart Failure](#) for detailed information on specific medications and classes.

Guideline-directed medical therapy of heart failure with reduced ejection fraction (HFrEF)

Guideline-directed medical therapy consists of evidence-based medications that have been shown to improve clinical outcomes for patients with HF. Medical treatment of chronic HFrEF generally consists of 4 foundational classes of therapy for all patients, as all classes have been demonstrated to reduce mortality ([1](#), [2](#), [4](#)). These include:

- Beta-blockers

- Angiotensin receptor/neprilysin inhibitor (ARNI)
- Mineralocorticoid receptor antagonist (MRA)
- Sodium-glucose co-transporter 2 (SGLT2) inhibitor

Doses of medications in these pharmacologic classes are generally titrated to maximum tolerable doses to achieve optimal benefit.

Beta-blockers (eg, [bisoprolol](#), [carvedilol](#), [metoprolol](#)) reduce sympathetic nervous system tone and overall neurohormonal activation and have been a mainstay of chronic heart failure therapy ([1](#), [2](#)). Beta-blockers can also help to control arrhythmias associated with heart failure, in addition to being antianginal agents for patients with ischemic cardiomyopathy. Reduction of basal heart rate has been demonstrated to be an important marker of outcome. Because of their negative inotropic effects, beta-blockers are avoided in the acute decompensated phase of heart failure; they are initiated when a patient is stable and well compensated. Beta-blockers are the only foundational class of pharmacologic chronic heart failure agents that have demonstrated value only for patients with reduced ejection fraction, and not for patients with preserved left ventricular ejection fraction. For patients who are unable to tolerate beta-blockers or who achieve suboptimal resting heart rate reduction (> 77 beats/minute for those in sinus rhythm), [ivabradine](#), a sinus node inhibitor, can be used to achieve heart rate reduction.

Treatment with an **angiotensin receptor/neprilysin inhibitor** ([sacubitril/valsartan](#)) is the standard for patients with chronic heart failure ([1](#), [2](#), [5](#), [6](#), [7](#)). This medication combination is a potent vasodilator and may induce symptomatic hypotension. Therefore, caution is recommended when initiating or uptitrating sacubitril/valsartan, especially for patients with a systolic blood pressure of less than 100 mm Hg. Sacubitril/valsartan has demonstrated superior mortality reduction, compared to angiotensin-converting enzyme-inhibitors or angiotensin II receptor blockers, although both classes of medication are still used routinely, particularly in patients who are intolerant of sacubitril/valsartan.

[Spironolactone](#) and [eplerenone](#) are **mineralocorticoid receptor antagonists** used for the treatment of chronic heart failure ([1](#), [2](#), [8](#)). Agents, such as [finerenone](#), are under investigation in different patient subpopulations, including those with significant diabetes and those with significant renal dysfunction. Spironolactone has a relatively weak diuretic effect, but is potassium sparing and can be associated with hyperkalemia. Potassium levels must be monitored in patients when initiating spironolactone or when increasing the dose. Spironolactone can also be associated with gynecomastia, and should be discontinued if this adverse effect develops. Eplerenone has a more favorable adverse effect profile, including less relative hyperkalemia and gynecomastia.

SGLT2 inhibitors (eg, [dapagliflozin](#), [empagliflozin](#)) have demonstrated significant survival benefit for patients with heart failure, regardless of ejection fraction ([1](#), [2](#), [9](#), [10](#)). They are generally well tolerated and have a relatively weak diuretic effect. Originally developed as oral hypoglycemic agents, they can reduce blood glucose levels.

Other pharmacologic therapies are available for specific populations of patients with chronic heart failure, particularly those who remain symptomatic despite optimal doses of the foundational pharmacologic therapy classes ([1](#), [2](#)).

- [Digoxin](#) can be used to reduce the risk of heart failure hospitalizations in patients with reduced ejection fraction. Digoxin is contraindicated in patients with moderate to severe chronic kidney disease due to the risk of toxicity.
- Additional diuretic therapy, particularly loop diuretics, are frequently used in patients with ongoing symptoms of congestion, despite optimal medical therapy for chronic heart failure.
- [Ivabradine](#) can be used to control resting heart rate in patients who are in sinus rhythm who are either poorly tolerant of beta-blockers or do not achieve satisfactory resting heart rate control with them. Ivabradine is recommended therapy for patients with HFrEF and an EF < 35% with a resting heart rate ≥ 70 beats per minute on beta blocker therapy ([11](#)).
- [Vericiguat](#) is an oral, soluble guanylate cyclase agonist that has vasodilator effects. It can be added to other heart failure therapies in patients with heart failure with reduced ejection fraction with ongoing symptoms who have also had an acute decompensation within the last 6 months ([12](#)).
- Vasodilators, as a combination of [isosorbide dinitrate](#) and [hydralazine](#), can be used in patients with persistent symptoms of congestion, particularly those who are unable to tolerate ARNIs or when they are contraindicated due to poor kidney function.

Calcium channel blockers, nonsteroidal anti-inflammatory drugs, thiazolidinediones, and class 1C antiarrhythmic medications can worsen heart failure and should be avoided unless no alternative exists; patients who must take such medications should be followed closely.

Medical therapy of heart failure with mildly reduced or improved ejection fraction (HRmrEF and HRimpEF)

In general, the therapeutic approach for HRmrEF and HRimpEF is similar to that for HFrEF, although far less evidence exists to support specific medications or classes ([1](#), [2](#)). Diuretics are recommended for anti-congestive therapy. SGLT2 inhibitors may be beneficial in preventing hospitalization in HRmrEF. Data directly supporting beta blockade, angiotensin-renin system inhibition, and mineralocorticoid antagonists are lacking and recommendations are based on observational or post-hoc analyses.

Medical therapy of heart failure with preserved ejection fraction (HFpEF)

Recommended therapies for HFpEF include an SGLT2 inhibitor for all patients, diuretics for patients with congestion and symptoms, and mineralocorticoid antagonists and angiotensin-renin system inhibition for select patient groups ([1](#), [2](#), [13](#)). Cardiovascular comorbidities, including hypertension, atherosclerotic disease, and atrial fibrillation should be treated. As with HRmrEF, SGLT2 inhibitors, beta blockers, angiotensin-renin system inhibitors, and mineralocorticoid antagonists may be considered.

See [Medications for Heart Failure](#) for more detailed information on specific medication classes.

Disease management

General measures, especially patient and caregiver education and diet and lifestyle modifications, are important for all patients with heart failure ([1](#)).

- Education

- Sodium restriction
- Appropriate weight and fitness levels
- Correction of underlying conditions

Patient and caregiver education are critical to long-term success. The patient and family should be involved in treatment choices. They should be taught the importance of medication adherence, warning signs of an exacerbation, and how to link cause with effect (eg, increased salt in the diet with weight gain or symptoms).

Many centers (eg, specialized outpatient clinics) have integrated health care practitioners from different disciplines (eg, HF nurses, pharmacists, social workers, rehabilitation specialists) into multidisciplinary teams or outpatient heart failure management programs. These approaches can improve outcomes and reduce hospitalizations and are most effective in the sickest patients.

Dietary sodium restriction helps limit fluid retention. All patients should eliminate salt in cooking and at the table and avoid salted foods; the most severely ill should limit sodium to < 2 g/day by consuming only low-sodium foods ([1](#), [2](#)).

Monitoring daily morning weight helps detect sodium and water accumulation early ([1](#), [2](#)). If weight increases > 2 kg over a few days, patients may be able to adjust their diuretic dose themselves, but if weight gain continues or symptoms occur, patients should seek medical attention.

Intensive case management, particularly by monitoring medication adherence and frequency of unscheduled visits to the physician or emergency department and hospitalizations, can identify when intervention is needed ([1](#), [2](#)). Nurses who specialize in the care of patients with HF are valuable in education, follow-up, and dosage adjustment according to predefined protocols.

Patients with atherosclerosis or diabetes should strictly follow a diet appropriate for their disorder. Obesity may cause and always aggravates the symptoms of HF; patients should attain a body mass index (BMI) ≤ 30 kg/m² (ideally 21 to 25 kg/m²).

Regular light activity (eg, walking), tailored to symptoms, is encouraged ([1](#), [2](#)). Activity prevents skeletal muscle deconditioning, which worsens functional status; however, activity does not appear to improve survival or decrease hospitalizations ([1](#), [2](#)). Rest is appropriate during acute exacerbations. Formal exercise through [cardiac rehabilitation](#) is useful for patients with chronic HFrEF and is likely helpful for patients with chronic HFpEF.

Patients should have annual [influenza vaccination](#) because influenza can precipitate HF exacerbations, particularly in institutionalized or older patients. Patients should be [vaccinated against SARS-CoV-2](#).

If causes such as hypertension, persistent tachyarrhythmia, severe anemia, hemochromatosis, uncontrolled diabetes, thyrotoxicosis, beriberi, alcohol use disorder, or toxoplasmosis are successfully treated, patients may dramatically improve.

Anemia is common among patients with chronic heart failure and is frequently multifactorial. Anemia is associated with worse symptoms and outcomes in HF and so reversible causes should be sought and treated. [Iron deficiency](#) is among the most common causes of anemia in HF, and iron replacement

therapy should be considered once treatable causes such as blood loss (gastrointestinal or other) have been excluded (2). Oral iron replacement is often less effective due to poor absorption and other reasons, thus intravenous iron replacement is preferred.

Management of extensive ventricular infiltration (eg, in amyloidosis) has improved considerably. [Treatments for amyloidosis](#) have markedly improved prognosis for many of these patients.

Device therapy

Use of an [implantable cardioverter-defibrillator](#) (ICD) or [cardiac resynchronization therapy](#) (CRT) is appropriate for some patients.

An **implantable cardioverter-defibrillator** is recommended for patients with an otherwise good life expectancy if they have symptomatic sustained ventricular tachycardia or ventricular fibrillation or if they remain symptomatic and have a left ventricular ejection fraction persistently < 35% while receiving guideline-directed medical therapy (2). The data for ICD use in HFrEF are stronger for ischemic cardiomyopathy than in nonischemic cardiomyopathy. A clinical trial that included patients with HFrEF with nonischemic cardiomyopathy demonstrated no mortality benefit from prophylactic (primary prevention) ICD placement (14).

Cardiac resynchronization therapy is a mode of pacing that synchronizes contraction of the left ventricle by simultaneously pacing its opposing wall, thereby improving stroke volume. CRT may relieve symptoms and reduce heart failure hospitalizations for patients who have HF, left ventricular ejection fraction < 35%, and a widened QRS complex with a left bundle branch block pattern (> 0.15 second—the wider the QRS, the greater potential benefit) (2). Many CRT devices also incorporate an ICD in their mechanism.

An implantable device that remotely monitors invasive hemodynamics (eg, pulmonary artery pressure) to guide therapy can help reduce frequency of heart failure exacerbation and hospitalization in selected patients (4, 15).

Durable or ambulatory **LV assist devices** (LVADs) are longer-term implantable pumps that augment LV output. They are commonly used to maintain patients with severe HF who are awaiting transplantation and are also used as "destination therapy" (ie, as a long-term or permanent solution) in some patients who are not candidates for transplant.

Surgery and percutaneous procedures

Surgery may be appropriate when certain underlying disorders are present. Surgery in patients with advanced HF should be done in a specialized center.

Surgical closure of [congenital](#) or acquired intracardiac shunts can be curative.

[Coronary artery bypass grafting](#) (CABG) for patients with LV systolic dysfunction secondary to coronary artery disease and evidence of myocardial viability may be beneficial (1, 2); however, those patients with prior myocardial infarction and non-viable myocardium are less likely to benefit from CABG. Thus, the

decision to use revascularization in a patient with HF and multivessel coronary artery disease should be made on a case-by-case basis.

If HF is primarily due to a [valvular disorder](#), valve repair or replacement should be considered ([1](#), [2](#)). Patients with primary mitral regurgitation are more likely to benefit than patients with mitral regurgitation secondary to LV dilation, in whom poor myocardial function is likely to continue postoperatively. Surgery is preferably done before myocardial dilation and damage become irreversible. Percutaneous mitral valve repair procedure (also called transcatheter end-to-end repair [TEER]), in which a clip is applied to approximate the anterior and posterior mitral leaflets, has been shown to reduce death and HF hospitalization in carefully selected patients with symptomatic HF despite optimal medical management and moderate to severe or severe mitral regurgitation with preserved LV size (end-systolic dimension ≤ 70 mm) ([16](#)).

[Heart transplantation](#) is the treatment of choice for patients < 70 years who have severe HF refractory to guideline-directed medical therapy ([1](#), [2](#)). Some older patients with otherwise good health are also considered if they meet other criteria for transplantation.

Treatment of persistent heart failure symptoms

Management may need to be adjusted when symptoms persist despite treatment. Reasons and appropriate actions include:

- Persistence of an underlying disorder (eg, hypertension, ischemia or infarction, valvular disease) despite treatment: Optimize treatment of underlying disorder
- Suboptimal treatment of heart failure: Escalate or optimize heart failure treatment regimen
- Medication nonadherence: Adherence strategies
- Excess intake of dietary sodium or alcohol: Reduction of sodium intake, alcohol intake, or both; refer to dietician or addiction specialist as necessary
- Presence of an undiagnosed thyroid disorder, anemia, or supervening arrhythmia (eg, atrial fibrillation with rapid ventricular response, intermittent ventricular tachycardia): Test for and treat undiagnosed disorders

End-of-life care

All patients and family members should be taught about disease progression and the risk of sudden cardiac death. For some patients, improving quality of life is as important as increasing quantity of life. Thus, it is important to determine patients' wishes about resuscitation (eg, [endotracheal intubation](#), CPR [cardiopulmonary respiration]) if their condition deteriorates, especially when HF is already severe.

All patients should be reassured that symptoms will be relieved, and they should be encouraged to seek medical attention early if their symptoms change significantly. Involvement of pharmacists, nurses, social workers, and clergy (when desired), who may be part of an interdisciplinary team or disease management program already in place, is particularly important in [end-of-life care](#).

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Prognosis for Chronic Heart Failure

HF usually involves gradual deterioration, interrupted by bouts of severe decompensation, and ultimately death, although the time course is being lengthened with modern therapies. However, death can also be sudden and unexpected, without prior worsening of symptoms. Overall 1-year mortality in patients with chronic HF is approximately 7 to 14% ([1](#), [2](#), [3](#)).

Specific factors that suggest a poor prognosis include older age at diagnosis, male gender, New York Heart Association class III or IV symptoms, ejection fraction < 30%, frequent or recent hospitalization, presence of chronic kidney or pulmonary disease, diabetes, lower measures of exercise tolerance (such as peak oxygen consumption or 6-minute walk distance), elevation of natriuretic peptide levels, and hyponatremia ([4](#), [5](#), [6](#)).

CLINICAL CALCULATORS

[MAGGIC Score Predicting Heart Failure Survival](#)



BNP, NTproBNP, and risk scores such as the [Meta-Analysis Global Group in Chronic Heart Failure \(MAGGIC\) Risk Score](#) and the [Seattle Heart Failure model](#), are helpful to predict prognosis in HF patients as an overall group, although there is significant variation in survival among individual patients ([7](#), [8](#)).

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Key Points

- Chronic heart failure affects over 6 million people in the United States.
- Chronic heart failure is staged based on structural and functional abnormalities as well as the presence and degree of symptoms.
- Pharmacologic treatment of chronic heart failure with reduced ejection fraction involves 4 key medication classes: beta-blockers, Angiotensin receptor/neprilysin antagonist, sodium-glucose co-transporter 2 (SGLT2) inhibitor, and mineralocorticoid receptor antagonist.
- Multidisciplinary care and lifestyle changes are important elements for treatment of chronic heart failure.
- Implantable cardioverter-defibrillators and cardiac resynchronization therapy are used in select patients.

Drug Information for the Topic



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Pulmonary Edema

Full Review: Oct 2025 By [Nowell M. Fine](#), MD, SM, Libin Cardiovascular Institute, Cumming School of Medicine, University of Calgary | Peer reviewed by [Jonathan G. Howlett](#), MD, Cumming School of Medicine, University of Calgary
Last updated: Oct 2025

Cardiogenic pulmonary edema is acute alveolar flooding caused by pulmonary venous hypertension secondary to severe left ventricular failure. Findings are severe dyspnea, diaphoresis, wheezing, and sometimes blood-tinged frothy sputum. Diagnosis is clinical and by chest radiograph. Treatment is with oxygen, IV nitrates, diuretics, and, in patients with heart failure and reduced ejection fraction, sometimes short-term IV positive inotropes and assisted ventilation (ie, endotracheal intubation with mechanical ventilation or bilevel positive airway pressure ventilation).

(See also [Acute Heart Failure](#).)

When left ventricular (LV) filling pressure increases suddenly, plasma fluid moves rapidly from pulmonary capillaries into interstitial spaces and alveoli, causing pulmonary edema. Although precipitating causes vary by age and country, 30 to 40% of cases result from [acute coronary ischemia](#) (1, 2). Some cases are from decompensation of significant underlying [heart failure](#) (HF), including HF with preserved ejection fraction (HFpEF) due to [hypertension](#), and the rest are from [arrhythmia](#), an acute [valvular disorder](#), or acute volume overload often due to IV fluids. Medication or dietary nonadherence is often involved.

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Symptoms and Signs of Pulmonary Edema

Patients with pulmonary edema present with extreme dyspnea, restlessness, and anxiety with a sense of suffocation. Cough, possibly producing blood-tinged sputum, pallor, cyanosis, and marked diaphoresis are common; some patients froth at the mouth. Frank hemoptysis is uncommon.

The pulse is rapid and low volume, and blood pressure (BP) is variable. Marked hypertension indicates significant cardiac reserve; hypotension with systolic BP < 100 mm Hg is ominous.

Inspiratory fine crackles are widely dispersed anteriorly and posteriorly over both lung fields. Marked wheezing may occur. Noisy respiratory efforts often make cardiac auscultation difficult; a summation gallop—merger of third (S3) and fourth (S4) heart sounds—may be present.

"Cardiac asthma" refers to coughing, wheezing, breathlessness, and sometimes paroxysmal nocturnal dyspnea, that occur with left heart failure and pulmonary edema.

Signs of right ventricular failure (eg, neck vein distention, peripheral edema) may be present.

Diagnosis of Pulmonary Edema

- History and physical examination showing severe dyspnea and pulmonary crackles
- Chest radiograph
- Sometimes chest ultrasound
- Sometimes arterial blood gas analysis
- Sometimes serum brain natriuretic peptide (BNP) or N-terminal-pro BNP (NT-pro-BNP)
- Additional [acute heart failure testing](#) as indicated

A [chronic obstructive pulmonary disease \(COPD\) exacerbation](#) can mimic pulmonary edema due to left ventricular failure. While a patient without any prior history of cardiac disease may present in new-onset heart failure with acute pulmonary edema, patients with severe dyspnea due to COPD usually have a history of COPD, although they may be too dyspneic to relate it.

Imaging in Patients With Pulmonary Edema

Chest Radiograph of a Patient with Pulmonary Edema

This radiograph shows bilateral perihilar patchy opacities in a "bat-wing" or "butterfly" pattern consistent with cardiogenic pulmonary edema. In this patient, the costophrenic angles are seen sharply, indicating the absence of significant pleural effusions.

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Chest Radiograph of a Patient with Pulmonary Edema and Bilateral Pleural Effusions

This patient has bilateral patchy interstitial infiltrates characteristic of pulmonary edema, as well as bilateral pleural effusions (arrows). The normally sharp costophrenic angles are obscured by fluid.

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Kerley B Lines

Kerley B lines (arrows) are horizontal lines in the lung periphery that extend to the pleural surface. They denote thickened, edematous interlobular septa often due to pulmonary edema.

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Chest Radiograph of a Patient with Cardiomegaly and Cephalization

This patient has cardiomegaly (width of cardiac silhouette is > 50% of thoracic cage on a posterior-anterior chest radiograph). The patient also has cephalization (black arrows) whereby upper lobe pulmonary vessels become more prominent. These findings are often seen in patients with heart failure.

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Chest Radiograph of a Patient with Fluid in Minor and Major Fissures

Lateral chest radiograph in a patient with fluid in major (1) and minor (2) fissures as well as a loculated effusion (3) seen along right posterior chest wall.

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Comet Tail Artifact

Sonographic "comet tail artifact" (white arrow) inferior to rib (black arrows) on thoracic ultrasound. Comet tail artifact corresponds to the radiographic finding of Kerley B lines.

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A **chest radiograph**, done immediately, is usually diagnostic, showing marked interstitial edema. **Chest ultrasound** may also be useful, particularly if pleural effusions are suspected.

Pulse oximetry and **arterial blood gas analysis** may reveal hypoxemia, hypercapnia, or metabolic acidosis if tissue perfusion is impaired. Hypoxemia can be severe. Carbon dioxide retention is a late, ominous sign of secondary hypoventilation. Bedside measurement of **serum BNP or NT-pro BNP** levels (elevated in pulmonary edema; normal in COPD exacerbation) is helpful if the diagnosis is in doubt.

ECG and other blood tests (cardiac biomarkers, electrolytes, blood urea nitrogen [BUN], creatinine) are performed.

Echocardiography is helpful to determine the cause of the pulmonary edema (eg, [myocardial infarction](#), [valvular dysfunction](#), [hypertensive heart disease](#), [dilated cardiomyopathy](#)), as well as evaluating the patient's overall hemodynamic status, and may influence the choice of therapies.

Treatment of Pulmonary Edema

- Treatment of cause
- Oxygen
- Ventilatory support as indicated
- IV diuretic
- Sometimes nitrates
- Sometimes IV inotropes

Treatment of pulmonary edema involves providing oxygen and appropriate ventilatory support, administering diuretics (often in combination with vasodilators and/or inotropes depending upon the blood pressure, perfusion, and cardiac function), identifying and correcting the underlying cause (including any arrhythmias), and optimizing overall medical therapy. (see figure [Management of Patients With Pulmonary Edema](#)).

Management of Patients With Pulmonary Edema

MCS = mechanical circulatory support; RRT = renal replacement therapy; SBP = systolic blood pressure.

Adapted from [McDonagh TA, Metra M, Adamo M, et al](#): 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure: Developed by the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of

Cardiology (ESC) with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur Heart J* 42(36):3599-3726, 2021. doi: 10.1093/eurheartj/ehab368.

[Morphine](#) has long been used to reduce severe anxiety and the work of breathing but is decreasingly used (except in palliative care) due to observational studies suggesting a poorer outcome with its use (1).

Noninvasive ventilatory assistance with bilevel positive airway pressure (BiPAP) is helpful if hypoxia is significant. If carbon dioxide retention is significant or the patient is obtunded, [tracheal intubation](#) and [mechanical ventilation](#) are indicated.

Specific additional treatment depends on etiology of pulmonary edema:

- For acute myocardial infarction or another acute coronary syndrome, thrombolysis or direct percutaneous coronary angioplasty with or without stent placement
- For severe hypertension, an IV vasodilator
- For supraventricular or ventricular tachycardia, direct-current cardioversion
- For rapid atrial fibrillation, cardioversion is preferred. To slow the ventricular rate, an IV beta-blocker, IV [digoxin](#), or cautious use of an IV calcium channel blocker

In patients with acute myocardial infarction, fluid status before onset of pulmonary edema is usually normal, so diuretics are less useful than in patients with acute decompensation of chronic heart failure and may precipitate hypotension. If systolic blood pressure falls < 100 mm Hg or shock develops, IV [dobutamine](#) and an intra-aortic balloon pump (counterpulsation) may be required.

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Key Points

- Acute pulmonary edema can result from acute coronary ischemia, decompensation of underlying heart failure, arrhythmia, an acute valvular disorder, or acute volume overload.
- Patients have severe dyspnea, diaphoresis, wheezing, and sometimes blood-tinged frothy sputum.
- Clinical examination and chest radiography are usually sufficient for diagnosis; ECG, cardiac markers, and sometimes echocardiography are done to identify cause.

- Treat the cause and give oxygen and IV **furosemide** and/or nitrates as needed; inotropic agents to increase cardiac output, try noninvasive ventilatory assistance initially but use tracheal intubation and assisted ventilation if necessary.

Drug Information for the Topic



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Right Heart Failure and Cor Pulmonale

Full Review: Oct 2025 By [Nowell M. Fine](#), MD, SM, Libin Cardiovascular Institute, Cumming School of Medicine, University of Calgary | Peer reviewed by [Jonathan G. Howlett](#), MD, Cumming School of Medicine, University of Calgary
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Right heart failure is defined by the clinical syndrome of systemic venous congestion with or without low cardiac output, secondary to right ventricular dysfunction. The most common cause of right heart failure is increased afterload caused by left heart failure. **Cor pulmonale** is right ventricular (RV) enlargement and eventual failure secondary to a lung disorder that causes pulmonary artery hypertension. Findings include peripheral edema, neck vein distention, hepatomegaly, and a parasternal lift. Diagnosis of right heart failure is clinical and by echocardiography and cardiac MRI. Treatment is directed at the cause.

Pathophysiology of Right Heart Failure

Right heart or right ventricular (RV) failure is the failure of the right side of the heart to pump blood adequately through the pulmonary vasculature at a normal central venous pressure. Right heart failure often involves right ventricle systolic dysfunction, dilation, or both. In the case of right heart failure secondary to left heart failure, elevated left heart filling pressure is transmitted retrograde through the pulmonary vasculature, resulting in pulmonary hypertension and right ventricular pressure overload, which can be exacerbated by the development of [tricuspid valve regurgitation](#). Tricuspid regurgitation is typically functional in the setting of either right ventricular enlargement, or right atrial enlargement, often exacerbated by the presence of atrial fibrillation. [Sleep apnea](#) is common in patients with [chronic heart failure](#), particularly central sleep apnea. The presence of both central and obstructive sleep apnea can exacerbate right heart failure.

Cor Pulmonale

Cor pulmonale is right heart failure caused by pulmonary hypertension secondary to lung disease, which can include:

- Loss of capillary beds (eg, due to bullous changes in [chronic obstructive pulmonary disease \[COPD\]](#) or thrombosis in [pulmonary embolism](#))

- Vasoconstriction caused by hypoxia, hypercapnia, or both
- Increased alveolar pressure (eg, in COPD, during [mechanical ventilation](#))
- Medial hypertrophy in arterioles (often a response to [pulmonary hypertension](#) due to other mechanisms)

Pulmonary hypertension increases afterload on the RV, resulting in a cascade of events that is similar to what occurs in LV failure, including elevated end-diastolic and central venous pressure and ventricular hypertrophy and dilation. Demands on the RV may be intensified by increased blood viscosity due to hypoxia-induced polycythemia. Rarely, RV failure affects the LV if a dysfunctional septum bulges into the LV, interfering with filling and thus causing diastolic dysfunction.

Cor pulmonale results from a disorder of the lung or its vasculature; it does not refer to right ventricular (RV) enlargement secondary to left ventricular (LV) failure, a congenital heart disorder (eg, [ventricular septal defect](#)), or an acquired valvular disorder. Cor pulmonale is usually chronic but may be acute and reversible. Primary [pulmonary hypertension](#) (ie, not caused by a pulmonary or cardiac disorder) is discussed elsewhere.

Etiology of Right Heart Failure and Cor Pulmonale

The most common cause of right heart failure is left heart failure (1). Isolated right heart failure can be caused by right-sided congenital or acquired [valvular disease](#), right-sided [myocardial infarction](#), [restrictive cardiomyopathy](#), [pericardial disease](#), right-heart specific cardiomyopathy such as [arrhythmogenic right ventricular cardiomyopathy](#), or [obstructive sleep apnea](#).

Cor pulmonale

Acute cor pulmonale has few causes. Chronic cor pulmonale is usually caused by [COPD](#), but there are several less common causes (see table [Causes of Cor Pulmonale](#)). In patients with COPD, an acute exacerbation or pulmonary infection may trigger RV overload. In chronic cor pulmonale, risk of [venous thromboembolism](#) is increased.

TABLE	
Causes of Cor Pulmonale	
Acuity	Condition
Acute	Massive pulmonary embolization
	Injury due to mechanical ventilation (most commonly in patients with ARDS)
Chronic	COPD *
	Extensive loss of lung tissue due to surgery or trauma

Acuity	Condition
	Chronic, unresolved pulmonary emboli
	Pulmonary veno-occlusive disorders
	Systemic sclerosis
	Pulmonary interstitial fibrosis
	Kyphoscoliosis
	Obesity with alveolar hypoventilation
	Neuromuscular disorders involving respiratory muscles
	Idiopathic alveolar hypotension
* COPD is the most common cause of chronic cor pulmonale. ARDS = acute respiratory distress syndrome; COPD = chronic obstructive pulmonary disease.	

Etiology reference

1. [Ezekowitz JA, O'Meara E, McDonald MA, et al.](#) 2017 Comprehensive Update of the Canadian Cardiovascular Society Guidelines for the Management of Heart Failure. *Can J Cardiol.* 2017;33(11):1342-1433. doi:10.1016/j.cjca.2017.08.022

Symptoms and Signs of Right Heart Failure and Cor Pulmonale

In RV failure, the most common symptoms are peripheral edema and fatigue. Sometimes patients feel a sensation of fullness in the abdomen or neck. Hepatic congestion can cause right upper quadrant abdominal discomfort, and stomach and intestinal congestion can cause early satiety, anorexia, and abdominal bloating.

Less specific heart failure symptoms include cool peripheries, postural light-headedness, nocturia, and decreased daytime micturition. Skeletal muscle wasting can occur in severe biventricular failure and may reflect some disuse but also increased catabolism associated with increased cytokine production. Significant weight loss (cardiac cachexia) is an ominous sign associated with high mortality.

In older adults, presenting concerns may be atypical, such as confusion, delirium, falls, sudden functional decline, nocturnal urinary incontinence, or sleep disturbance. Coexisting cognitive impairment and depression may also influence assessment and therapeutic interventions and may be worsened by the HF.

Signs of RV failure include:

- Nontender peripheral pitting edema (digital pressure leaves visible and palpable imprints, sometimes quite deep) in the feet and ankles
- Enlarged and sometimes pulsatile liver palpable below the right costal margin
- Abdominal swelling and ascites
- Visible elevation of the jugular venous pressure, sometimes with large *a* or *v* waves that are visible even when the patient is seated or standing (see figure [Normal Jugular Vein Waves](#))

In severe cases of heart failure, peripheral edema can extend to the thighs or even the sacrum, scrotum, lower abdominal wall, and occasionally even higher. Severe edema in multiple areas is termed anasarca. Edema may be asymmetric if patients lie predominantly on one side.

Large *v* waves in the jugular veins are usually indicative of significant tricuspid regurgitation which is often present in RV failure. A paradoxical increase in the jugular venous pressure during inspiration (Kussmaul sign) is indicative of impaired right ventricular filling and can occur in severe RV failure as well as [restrictive cardiomyopathy](#), [constrictive pericarditis](#), and severe [tricuspid regurgitation](#).

With hepatic congestion, the liver may be palpably enlarged or tender, and hepatojugular or abdominal-jugular reflux may be detected (see [Approach to the Cardiac Patient](#)). Precordial palpation may detect the left parasternal lift of RV enlargement, and auscultation may detect the murmur of tricuspid regurgitation or the RV third heart sound (S3) along the left sternal border; both findings are augmented upon inspiration.

Cor pulmonale

Cor pulmonale is initially asymptomatic, although patients usually have significant symptoms (eg, dyspnea, exertional fatigue) due to the underlying lung disorder. Later, as RV pressures increase, physical signs commonly include a left parasternal systolic lift, a loud pulmonic component of the second heart sound (S2), and murmurs of functional tricuspid and pulmonic insufficiency (regurgitation). Later, an RV gallop rhythm (S3 and fourth [S4] heart sounds) augmented during inspiration, distended jugular veins (with a dominant *a* wave unless tricuspid regurgitation is present), hepatomegaly, and lower-extremity edema may occur.

Diagnosis of Right Heart Failure and Cor Pulmonale

- Clinical suspicion
- Echocardiography and cardiac MRI
- Electrocardiogram
- Measurement of natriuretic peptide levels
- Sometimes right heart catheterization

Right heart failure is a clinical diagnosis, composed of supportive symptoms and physical examination findings. Cardiac imaging, particularly echocardiography and cardiac MRI, is used to confirm right

ventricular dysfunction, including dilatation and impaired systolic function. Both imaging modalities can identify previously undiagnosed structural, functional, and congenital abnormalities. Right atrial enlargement and functional tricuspid valve regurgitation are also supportive findings.

Echocardiography can assess RV systolic pressure but is often technically limited by the lung disorder; Cardiac MRI can quantitatively assess RV size and function. Right heart catheterization may be required for confirmation. Electrocardiography can identify arrhythmia as well as subtle findings of right ventricle hypertrophy and dilation and right atrial or ventricular dilation. Measurement of brain (B-type) natriuretic peptide (BNP) or N-terminal-pro-BNP (NT-pro-BNP) levels can support the diagnosis of heart failure as well as track clinical response over time.

Cor pulmonale

Cor pulmonale should be suspected in all patients with one of its causes. Chest radiograph shows RV and proximal pulmonary artery enlargement with distal arterial attenuation. ECG evidence of RV hypertrophy (eg, right axis deviation, QR wave in lead V1, and dominant R wave in leads V1 to V3) correlates well with degree of pulmonary hypertension. However, because pulmonary hyperinflation and bullae in COPD cause realignment of the heart, physical examination, radiographs, and ECG may be relatively insensitive.

Cor Pulmonale (Echocardiogram)

IMAGE

Short-axis view demonstrating a markedly enlarged right ventricle (RV) with RV hypertrophy. Abnormal bowing of the interventricular septum into the left ventricle (LV) gives a characteristic D configuration of the LV, consistent with volume and pressure overload of the RV.

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Treatment of Right Heart Failure and Cor Pulmonale

- Treatment of cause

Treatment of right heart failure consists of identifying and treating the underlying cause. When the cause is left heart failure, guideline-directed treatment strategies for left heart failure apply. [Pulmonary arterial hypertension](#), [congenital heart disease](#), [acquired valvular disease](#), [pericardial disease](#), [cardiomyopathy](#), and other causes are treated per standard of care.

In patients with acute or critical illness, careful attention to preload, as well as afterload and contractility are the main approaches (1). Cautious diuresis, inotropic support with agents such as [milrinone](#), and pulmonary vasodilation (in the absence of contraindications) may be helpful.

Symptoms may be managed with loop diuretics. Otherwise, there are no evidence-based targeted pharmacologic or other forms of therapies (2).

If peripheral edema is present chronically, diuretics may seem appropriate, but they should be used cautiously because small decreases in preload may result in hypotension. Pulmonary vasodilators (eg, [hydralazine](#), calcium channel blockers, nitrous oxide, prostacyclin analogs, phosphodiesterase inhibitors), although beneficial in select Group 1 (primary) pulmonary arterial hypertension patients, are generally not effective for other causes of cor pulmonale. Intravenous or inhaled agents are sometimes used in the acute setting in the absence of left heart dysfunction. [Bosentan](#), an endothelin receptor blocker, also may benefit patients with primary pulmonary hypertension, but its use is not well studied in cor pulmonale. [Digoxin](#) may be used empirically, however evidence is lacking in the absence of LV dysfunction.

Arrhythmias should also be managed.

See [Medications for Heart Failure](#) for more detailed information on specific medication classes.

Cor pulmonale

Treatment of cor pulmonale is difficult; it focuses on the cause (see table [Causes of Heart Failure](#)), particularly alleviation or moderation of hypoxia. Early identification and treatment are important before structural changes become irreversible.

Pulmonary arterial hypertension-specific therapies, including pulmonary vasodilators, are not typically indicated for cor pulmonale due to chronic lung disease, although they may play a role in severe cases (3).

Treatment references

1. [Houston BA, Brittain EL, Tedford RJ](#). Right Ventricular Failure. *N Engl J Med*. 2023;388(12):1111-1125. doi:10.1056/NEJMra2207410
2. [Ezekowitz JA, O'Meara E, McDonald MA, et al](#). 2017 Comprehensive Update of the Canadian Cardiovascular Society Guidelines for the Management of Heart Failure. *Can J Cardiol*. 2017;33(11):1342-1433. doi:10.1016/j.cjca.2017.08.022
3. [Blanco I, Hernández-González F, García A, Torres-Castro R, Barberà JA](#). Management of Pulmonary Hypertension Associated with Chronic Lung Disease. *Semin Respir Crit Care Med*.

Key Points

- Diagnosis usually requires echocardiography or radionuclide imaging, and sometimes right heart catheterization.
- Early identification and treatment of the cause are important before cardiac structural changes become irreversible.
- Loop diuretics are used cautiously for patients with symptoms or peripheral edema.
- Cor pulmonale is RV enlargement and eventually failure secondary to a lung disorder that causes pulmonary artery hypertension.
- Cor pulmonale itself is usually asymptomatic but common physical findings include a left parasternal systolic lift, a loud pulmonic component of S2, functional tricuspid and pulmonic regurgitation murmurs, and later, distended jugular veins, hepatomegaly, and lower-extremity edema.

Drug Information for the Topic



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Medications for Heart Failure

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Heart failure is a clinical syndrome in which the heart is not able to meet the metabolic demands of the body due to a structural and or functional cardiac abnormality, leading to low cardiac output, elevated ventricular filling pressure, or both (see [Heart Failure](#)).

The primary medications for long-term management and improved survival in patients with heart failure, particularly heart failure with reduced ejection fraction, are:

- [Beta-blockers](#)
- [Angiotensin receptor/neprilysin inhibitors \(ARNIs\)](#) (or [angiotensin converting enzyme \[ACE\] inhibitors](#) or [Angiotensin II receptor blockers \[ARBs\]](#))
- [Sodium-glucose cotransporter-2 inhibitors \(SGLT2i\)](#)
- [Mineralocorticoid receptor antagonists](#)

Additional medication classes used in patients with heart failure include:

- [Digoxin](#)
- [Diuretics](#)
- [Inotropes](#)
- [Sinus node inhibitors](#)
- [Soluble guanylate cyclase stimulator](#)
- [Vasodilators \(nitrates\)](#)

All patients should be given clear and explicit information about their medications, including the following:

- The importance of timely prescription renewal
- The importance of adherence to therapy
- How to recognize adverse effects
- When to contact their physician

(See also [Acute Heart Failure - Treatment](#), [Chronic Heart Failure - Treatment](#), and [Right Heart Failure](#) for more information about medication selection.)

Classes of Medications for Heart Failure

Beta-blockers

Beta-blockers act by competitively inhibiting beta-adrenergic receptors. [Bisoprolol](#) and [metoprolol](#) are considered cardioselective beta-blockers, primarily acting on beta-1 adrenergic receptors. [Carvedilol](#), a nonselective beta-blocker, is also a vasodilator with alpha-blocking and antioxidant effects. Beta-blockers produce a negative inotropic and chronotropic effect that reduces myocardial oxygen demand, modestly reduces left ventricular afterload, and moderates the release of catecholamines in response to stress.

In patients with **heart failure with reduced ejection fraction (HFrEF)**, beta-blockers, unless otherwise contraindicated (by [asthma](#), second-degree or third-degree [atrioventricular block](#), or previous significant intolerance), are a mainstay of treatment ([1](#), [2](#)). In acute decompensated HFrEF, beta-blockers are best started once the patient has stabilized and has no evidence of pulmonary congestion. For patients with HFrEF and acute heart failure exacerbation who are already taking a beta-blocker, the dose should not be decreased or stopped unless necessary. Specific beta-blockers such as [carvedilol](#) and [metoprolol](#) succinate (ie, long-acting [metoprolol](#)) improve left ventricular ejection fraction, survival, and other major cardiovascular outcomes in patients with chronic HFrEF, including those with severe symptoms. After initial treatment, heart rate and myocardial oxygen consumption decrease, and stroke volume and filling pressure are unchanged. With the slower heart rate, diastolic function improves. Ventricular filling returns to a more normal pattern (increasing in early diastole), which appears less restrictive. Improved myocardial function is measurable in some patients after 6 to 12 months but may take longer; ejection fraction (EF) and cardiac output increase, and LV filling pressure decreases. Exercise capacity improves.

In patients with **heart failure with preserved ejection fraction (HFpEF)**, beta-blockers are not generally recommended, although they can be used for rate control in atrial fibrillation with rapid ventricular response, or for control of hypertension in patients with coronary artery disease or prior myocardial infarction ([1](#), [2](#)).

The starting dose should be low (one-fourth of the target daily dose), then the dose is gradually increased over 8 weeks as tolerated. The acute negative inotropic effects of beta-blockade may initially cause cardiac depression and fluid retention. In such cases, a temporary increase in diuretic dose and slower upward titration of the beta-blocker dose is warranted. Tolerance may improve over time, and efforts should be made to reach target doses. Usual oral target doses are [carvedilol](#) 25 mg twice a day (50 mg twice a day for patients $\geq$ 85 kg), [bisoprolol](#) 10 mg once a day, or [metoprolol](#) succinate (extended-release) 200 mg once a day.

Renin-angiotensin system inhibiting medications

Angiotensin receptor/neprilysin inhibitors (ARNIs)

ARNIs are a combination medication for the treatment of heart failure. They include an ARB and a neprilysin inhibitor (eg, sacubitril). The ARB portion blocks the renin-angiotensin-aldosterone axis at the level of the angiotensin II receptor. Neprilysin is an enzyme involved in the breakdown of vasoactive

substances such as brain (B-type) natriuretic peptide (BNP) and other peptides. By inhibiting the breakdown of BNP and other beneficial vasoactive peptides, these medications lower blood pressure, decrease afterload, and enhance natriuresis. Because neprilysin inhibitors increase BNP levels, NTproBNP levels (which are not increased by the medication) should be used instead to help diagnose and manage heart failure.

In patients with **HFrEF**, the ARNI sacubitril/valsartan reduces all-cause mortality and should be considered in all patients with HFrEF ([1](#), [2](#)). It is preferred over an ACE inhibitor or ARB if tolerated, and evidence supports early transition of patients from an ACE inhibitor or ARB to ARNI, even in the hospital setting. In **patients with HFpEF**, [sacubitril/valsartan](#) may be considered, particularly in specific subgroups (women and patients with left ventricular ejection fraction below 57%).

The starting dose of [sacubitril/valsartan](#) is 49/51 mg orally twice a day for patients previously taking an ACE inhibitor or ARB, and 24/26 mg for patients previously taking a low dose of an ACE inhibitor or ARB (eg, ≤ 10 mg [enalapril](#) daily) or in those patients who have not taken an ACE inhibitor or an ARB or who have low or borderline blood pressure. ACE inhibitors should be discontinued 36 hours before initiation of sacubitril/valsartan. Patients previously taking an ARB can simply switch to sacubitril/valsartan without a washout period.

Complications associated with use of ARNI include hypotension, [hyperkalemia](#), renal insufficiency, and [angioedema](#). Sacubitril is coupled with [valsartan](#) (an ARB) because of the increased risk of angioedema with the use of sacubitril alone or in combination with an ACE inhibitor. For this reason, combined ACE inhibitor plus ARNI therapy is absolutely contraindicated.

Angiotensin converting enzyme (ACE) inhibitors

All **patients with HFrEF** should be given oral ACE inhibitors unless contraindicated.

ACE inhibitors reduce production of angiotensin II and breakdown of bradykinin, mediators that affect the sympathetic nervous system, endothelial function, vascular tone, and myocardial performance.

Hemodynamic effects include:

- Arterial and venous vasodilation
- Sustained decreases in LV filling pressure during rest and exercise
- Decreased systemic vascular resistance
- Favorable effects on ventricular remodeling

ACE inhibitors prolong survival and reduce hospitalizations due to heart failure ([1](#), [2](#)). In patients with **HFrEF**, their current role is as a second-line agent for those who cannot tolerate ARNi or those for whom ARNi is not available, cost-prohibitive, or otherwise unfeasible. For patients with **HFpEF**, ACE inhibitors are not typically initiated unless they are already being used for other indications such as [atherosclerosis](#), [hypertension](#), and [diabetic nephropathy](#).

The starting dose typically should be low (usually one-fourth to one-half of the target dose depending on blood pressure and renal function); the dose is gradually adjusted upward over 8 weeks as tolerated,

then continued indefinitely. Usual target doses of representative medications include [enalapril](#) 10 to 20 mg twice a day, [lisinopril](#) 20 to 30 mg once a day, and [ramipril](#) 5 mg twice a day; there are many others.

Serum electrolytes and renal function should be measured before an ACE inhibitor is initiated, at 1 month, and after each significant increase in dose or change in clinical condition. If dehydration or poor renal function due to acute illness develops, the ACE inhibitor dose may need to be reduced or temporarily discontinued. Contraindications include plasma creatinine > 2.8 mg/dL [> 250 micromol/L], bilateral [renal artery stenosis](#), renal artery stenosis in a solitary kidney, or previous angioedema due to ACE inhibitors.

The hypotensive effect (more marked in patients with [hyponatremia](#) or volume depletion), if troublesome, can often be minimized by separating administration of other blood pressure-lowering medications, reducing the dose of concomitant diuretics, using a longer acting ACE inhibitor (eg, [perindopril](#)), or giving the dose at bedtime. ACE inhibitors often cause mild to moderate reversible serum creatinine elevation due to vasodilation of the efferent glomerular arteriole. An initial 20 to 30% increase in creatinine is not a reason to stop the medication but does require closer monitoring, slower increases in dose, reduction in diuretic dose, or avoidance of nonsteroidal anti-inflammatory drugs (NSAIDs). Because aldosterone's effect is reduced, potassium retention ([hyperkalemia](#)) may result, especially in patients receiving potassium supplements. Cough occurs in many patients, probably because bradykinin accumulates, but other causes of cough should also be considered. If cough limits use, an ARB is a reasonable alternative. Occasionally, rash or dysgeusia occurs. [Angioedema](#) is rare but can be life threatening and is a contraindication to ACE inhibitors. Alternatively, ARBs can be used, although rarely cross-reactivity is reported. Both are contraindicated in pregnancy.

Angiotensin II receptor blockers (ARBs)

ARBs target the angiotensin II receptor rather than the angiotensin converting enzymes. They are less likely to cause cough and angioedema than ACE inhibitors, and may be used when these adverse effects prohibit ACE inhibitor use (Heidenreich).

In patients with **HF_{rEF}**, ACE inhibitors and ARBs are likely equally effective. In patients with **HF_{pEF}**, ARBs (as with ACE inhibitors) are not typically initiated unless they are already being used for other indications such as hypertension, diabetic kidney disease, or microalbuminuria.

Usual oral target doses are [valsartan](#) 160 mg twice a day, [candesartan](#) 32 mg once a day, and [losartan](#) 50 to 100 mg once a day. Introduction, upward dose adjustment, and monitoring of ARBs and ACE inhibitors are similar. Like ACE inhibitors, ARBs can cause reversible renal dysfunction, and the dose may need to be reduced or stopped temporarily during an acute dehydrating illness.

Adding an ARB to a regimen of an ACE inhibitor, beta-blocker, and mineralocorticoid receptor antagonist is unlikely to be helpful and should be avoided given the risk of [hyperkalemia](#). If a patient who is taking an ACE inhibitor or ARB still has symptoms, a mineralocorticoid receptor antagonist should be started, and/or an angiotensin receptor/neprilysin inhibitor (ARNI) should be used.

ARBs are contraindicated in pregnancy.

Mineralocorticoid receptor antagonists (aldosterone antagonists)

Mineralocorticoid receptor antagonists block aldosterone receptors in the distal convoluted tubule and collecting ducts of the kidney. They are considered potassium-sparing diuretics due to their relatively weak diuretic effect and the fact that, unlike most other diuretics, they do not waste serum potassium. For patients with heart failure, these agents offer blockade of the renin-angiotensin-aldosterone system that is complementary to the mechanism of ACE inhibitors and ARBs.

A mineralocorticoid receptor antagonist (MRA, also termed aldosterone antagonists) is recommended for patients with **HFrEF**, particularly those with symptoms (New York Heart Association Class II to IV) ([1](#), [2](#)). In patients with **HFpEF**, MRAs appear to have the most benefit as primary heart failure therapy in patients at the lower end of the ejection fraction spectrum and are also routinely used in patients with preexisting [hypertension](#) or [coronary artery disease](#).

Typical medications include [spironolactone](#) 25 to 50 mg orally once a day and [eplerenone](#) (which does not cause gynecomastia in males as spironolactone can). Mineralocorticoid receptor antagonists can reduce mortality, including from sudden death, in patients with left ventricular ejection fraction (LVEF) < 30% and chronic heart failure, or acute heart failure complicating [acute myocardial infarction](#).

Potassium supplements should be stopped. Serum potassium and creatinine should be checked every 1 to 2 weeks for the first 4 to 6 weeks and after dose changes. Dose is lowered if potassium is between 5.0 and 5.5 mEq/L (5.5 mmol/L) and stopped if potassium is > 5.5 mEq/L (5.5 mmol/L), if creatinine increases above 2.5 mg/dL (220 micromol/L), or if ECG changes of hyperkalemia are present.

Sodium-glucose cotransporter-2 inhibitors (SGLT2i)

SGLT2i were originally used in treatment of diabetes to block glucose reabsorption, thus causing glycosuria and lowering plasma glucose; they may also have effects on the myocardium and vasculature. These medications had previously been shown to prevent the onset of heart failure in patients with type 2 diabetes. The SGLT2i [dapagliflozin](#) and [empagliflozin](#) were subsequently shown to reduce cardiovascular mortality and hospitalization in patients with **HFrEF** who are symptomatic (New York Heart Association Class II to IV). Thus, SGLT2i therapy is recommended for all **HFrEF** patients with symptoms. SGLT2i therapy also appears beneficial in patients with **HFpEF**.

[Dapagliflozin](#) and [empagliflozin](#) may be given 10 mg orally once a day. With treatment, there is a mild (10 to 15%) reduction in estimated glomerular filtration rate (eGFR) which does not progress, glucosuria, and a small reduction in body weight. Risks include genital fungal infection, and in patients with diabetes, a very small risk of hypoglycemia and diabetic ketoacidosis. These medications are generally not indicated in patients with type I diabetes, low blood pressure, or low eGFR (< 30 mL/min/1.73 m²).

Digoxin

[Digoxin](#) inhibits the sodium-potassium pump (Na⁺, K⁺-ATPase), causing weak positive inotropy, reduces sympathetic activity, blocks the atrioventricular node (slowing the ventricular rate in atrial fibrillation or prolonging the PR interval in sinus rhythm), reduces vasoconstriction, and improves renal blood flow.

Digoxin has no proven survival benefit but, when used in addition to standard guideline-directed medical therapy, may help control symptoms and reduce the likelihood of hospitalization in patients with **HFrEF** (1, 2). Digoxin may also be used for rate control in patients with [atrial fibrillation](#) and both HFrEF and **HFpEF**.

In patients with normal renal function, digoxin, 0.125 to 0.25 mg orally once a day depending on age, sex, and body size, achieves full digitalization in about 1 week (5 half-lives). More rapid digitalization can be achieved but, unlike in the treatment of atrial fibrillation, there is typically little reason to rapidly digitalize (ie, digoxin load) patients with heart failure. Thus, simply starting digoxin at 0.125 mg orally once a day (in patients with normal renal function) or digoxin 0.125 mg orally every Monday, Wednesday, and Friday (in patients with abnormal renal function) is sufficient in patients with heart failure.

Digoxin toxicity is a concern given its narrow therapeutic window, especially in patients with renal dysfunction and perhaps in women. These patients may need a lower oral dose, as may older patients, patients with a low lean body mass, and patients also taking [amiodarone](#). The most important toxic effects are life-threatening arrhythmias (eg, [ventricular fibrillation](#), [ventricular tachycardia](#), complete [atrioventricular block](#)). Bidirectional ventricular tachycardia, nonparoxysmal junctional tachycardia in the presence of atrial fibrillation, and hyperkalemia are serious signs of digitalis toxicity. Nausea, vomiting, anorexia, diarrhea, confusion, amblyopia, and, rarely, xerophthalmia may also occur. If [hypokalemia](#) or [hypomagnesemia](#) (often due to diuretic use) is present, lower doses and serum levels can still cause toxicity; thus, electrolyte levels should be monitored in patients taking diuretics and digoxin.

Diuretics

While [spironolactone](#) is recommended for all patients with **HFrEF** and used for many with **HFpEF**, additional diuretics are given to patients with heart failure (regardless of underlying ejection fraction) who have current or previous volume overload; dose is adjusted to the lowest dose that stabilizes weight and relieves symptoms (1, 2). Chronic diuretics may be scheduled or used as needed by patients based on peripheral edema or changes in weight. Diuretics are a primary therapy in symptomatic **HFpEF**. They are used with caution in **right heart failure**, particularly in cor pulmonale, due to sensitivity to small changes in preload.

Loop diuretics should be used initially for control of volume overload, but their dose should be reduced when possible in favor of [mineralocorticoid receptor antagonists](#).

Thiazide diuretics are not normally used alone unless being given as treatment of hypertension; however, a thiazide diuretic may be added to a loop diuretic for additional diuresis and to reduce the loop diuretic dose. [Hydrochlorothiazide](#), [metolazone](#), and [chlorthalidone](#) can be used in this manner.

Commonly used loop diuretics include [furosemide](#), [bumetanide](#), and [torsemide](#). The starting dose of these medications depends on whether the patient has previously received loop diuretics. Common starting doses are:

- Furosemide 20 to 40 mg orally once a day or twice a day

- Bumetanide 0.5 to 1.0 mg orally once a day
- Torsemide 10 to 20 mg orally once a day

If needed, loop diuretics can be titrated up to doses of furosemide 120 mg orally twice a day, bumetanide 2 mg orally twice a day, and torsemide 40 mg orally twice a day based on response and renal function. Bumetanide and torsemide have better bioavailability than furosemide. If patients are switched between different loop diuretics, they should be placed on equivalent doses. Furosemide 40 mg is equivalent to bumetanide 1 mg, and both are equivalent to torsemide 20 mg.

In refractory cases, IV loop diuretics or [metolazone](#) 2.5 to 10 mg orally can be used for an additive effect. IV infusion of [furosemide](#) (5 to 10 mg/hour) or other loop diuretics may be helpful in selected patients with severe pulmonary edema. A bolus dose of loop diuretic should be given before starting an IV infusion and before each increase in infusion rate.

Loop diuretics (particularly when used in combination with thiazide diuretics) may cause hypovolemia with hypotension, hyponatremia, hypomagnesemia, and severe hypokalemia. The dose of diuretic required acutely can usually be gradually reduced; the target is the lowest dose that maintains stable weight and controls symptoms. When heart failure improves, the diuretic may be stopped if other medications improve heart function and relieve symptoms of heart failure. Using diuretic doses that are larger than required lowers cardiac output, impairs renal function, causes hypokalemia, and increases mortality. Serum electrolytes and renal function are monitored, initially daily (when diuretics are given IV) and subsequently as needed, particularly after a dose increase.

Vasopressin (antidiuretic hormone) receptor antagonists are not frequently used though they may be helpful in cases of severe refractory hyponatremia in patients with heart failure.

Inotropes

Various **positive inotropic medications** have been evaluated in heart failure but, except for [digoxin](#), they increase mortality risk. These medications can be grouped by mode of action as:

- Adrenergic (norepinephrine, epinephrine, dobutamine, dopamine)
- Nonadrenergic (enoximone and milrinone [phosphodiesterase type 3 inhibitors], levosimendan [calcium sensitizer])

Inotropic support is important for treatment of **acute heart failure** with cardiogenic shock. The choice of inotrope depends on availability, local practice, and afterload. For example, in a patient with severely reduced LV ejection fraction, peripheral vasoconstriction, and high or normal blood pressure, initial inotropic support may be combined with an IV vasodilator such as [nitroprusside](#), and then transitioned to [milrinone](#) (which has mild vasodilatory effects). On the other hand, a patient with reduced ejection fraction and hypotension with peripheral vasodilation may require an inotrope with vasopressor effects ([norepinephrine](#), high dose [epinephrine](#)), or a combination of an inotrope and a pure vasoconstricting agent. Synergy can be obtained by using both an adrenergic and nonadrenergic inotrope.

In patients with advanced (Stage D) **HFrEF** refractory to standard therapies, long-term continuous inotropic infusions may be used as a bridge to transplantation or as palliative therapy (2).

Sinus node inhibitors

[Ivabradine](#) is an I_f channel (inward "funny" channel in the sinus node) blocker that acts at the sinus node to slow the heart rate. Since these channels are present primarily in cardiac pacemaker cells, these medications are not useful for treatment in patients who are not in sinus rhythm. Ivabradine is recommended for use in patients with **HF_{rEF}** who have symptomatic heart failure (New York Heart Association Class II or III), normal sinus rhythm, and heart rate > 70 beats/minute despite guideline-directed medical therapy (at target beta-blocker dose or cannot tolerate a further increase in beta-blocker dose) ([1](#), [2](#)).

Initial dose of [ivabradine](#) is 2.5 to 5 mg orally twice a day, titrated at 2-week intervals to a heart rate of 50 to 60 beats/minute; maximum dose is 7.5 mg twice a day.

Soluble guanylate cyclase stimulator

[Vericiguat](#) is an oral soluble guanylate cyclase stimulator that enhances the cyclic guanosine monophosphate (GMP) pathway and sensitizes soluble guanylate cyclase to endogenous nitric oxide, resulting in pulmonary vasodilation. Vericiguat is an option to improve outcomes for patients with **HF_{rEF}** with worsening heart failure symptoms despite maximal guideline-directed medical therapy, with the potential to reduce mortality or hospitalization.

Vasodilators

[Hydralazine](#) and nitrates (such as [isosorbide dinitrate](#)) are direct vasodilators, which can improve hemodynamics, reduce valvular regurgitation, and increase exercise capacity without causing significant renal impairment.

The combination of [hydralazine](#) and [isosorbide dinitrate](#) may be considered in patients with **HF_{rEF}** who are intolerant of ACE inhibitors or ARBs (usually because of significant renal dysfunction) ([1](#), [2](#)). Additionally, in patients of African ancestry, this combination, when added to standard therapy, may reduce mortality and hospitalization, and improve quality of life.

When used instead of ACE inhibitor or ARB therapy, [hydralazine](#) is started at 25 mg orally 4 times a day and increased every 3 to 5 days to a target total dose of 300 mg/day; however, many patients cannot tolerate > 200 mg/day because of hypotension. [Isosorbide dinitrate](#) is started at 20 mg orally 3 times a day (with a 12-hour nitrate-free interval) and increased to a target of 40 to 50 mg 3 times a day. When added to ACE inhibitor or ARB therapy, hydralazine plus nitrate therapy is started at 37.5 mg and [isosorbide dinitrate](#) at 20 mg orally 3 times a day, with the maximum dose 75 mg and 40 mg 3 times a day. These doses are also available as a fixed-dose combination.

Nitrates and phosphodiesterase-5 inhibitors have been used to relieve symptoms in patients with **HF_{pEF}**, but neither is recommended for routine use for these patients ([2](#)).

Medication classes references

1. [McDonagh TA, Metra M, Adamo M, et al](#): 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure: Developed by the Task Force for the diagnosis and treatment of

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2. [Heidenreich PA, Bozkurt B, Aguilar D, et al](#): 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure: A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation* 145:e895–e1032, 2022, doi: 10.1161/CIR.0000000000001063

Drug Information for the Topic



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